Unit 2

Lesson 2

Innovators in Action.

Glove on Paper.

Time Frame: [80-90 min].

Ready, Set, Learn!

By the end of this lesson students will be able to:

- Use grid mapping and coordinates to plan and sketch a wearable glove circuit before building it..
- Identify cultural perspectives on the use of wearable technology.
- Evaluate a glove design for comfort, clarity, and usefulness.

Challenge questions of the lesson

- Why might different cultures design wearable technology differently?
- How would grid mapping help you plan the layout of a glove?
- What features would make a wearable glove design comfortable and easy to use?
- How can understanding cultural perspectives improve the design of wearable technology

SEL Relation

- **Self-Awareness:** Students reflect on their design choices and physical movements, developing a better understanding of their own abilities and creativity.
- **Fostering Social Awareness:** Through peer review and cultural research, students learn to appreciate diverse perspectives and cultural influences on technology.
- **Enhancing Relationship Skills:** Collaboration in building, testing, and providing constructive feedback strengthens communication, teamwork, and respect.
- **Encouraging Responsible Decision-Making:** Iterative design and reflection cultivate problem-solving skills and thoughtful decision-making in real-world contexts.

Engage

Teacher's Role:

- Read the story aloud with enthusiasm, emphasizing key ideas about designing the glove and using copper tape to send signals.
- Use visuals (like a paper glove outline or sample copper tape) to support understanding.
- Pause periodically to ask questions and check comprehension.
- Facilitate the “Let’s Think” activity by reading the question aloud and encouraging students to consider comfort, materials, and functionality.
- Provide examples or prompt students to think about materials they have used or seen in gloves and wearable tech.
- Support students who need extra help expressing their ideas by offering sentence starters or visuals.
- Encourage respectful sharing of ideas in the class discussion.

Expected from Students:

- Listen attentively to the story and visualize the glove design process.
- Ask clarifying questions if something is unclear.
- Reflect on the “Let’s Think” question and write or share ideas about materials and design considerations.
- Participate actively and respectfully in class discussions.
- Demonstrate curiosity and creativity in thinking about glove design.

Parts of the book included.


Pyramakerz

(Adam and Laila are sitting at their work table with a paper glove outline, copper tape, and a marker.).

Adam: “Okay, let’s figure out where we should put the copper tape.”

Laila: “The tape needs to touch when two fingers tap, right? That’s how it sends the signal.”

Adam: “Right! Let’s start with the index finger and thumb, and connect those to the Arduino.”

Laila: “Should we add more fingers too? Each pair could send a different message.”

Adam: “Like a secret code! Each finger tap means something new.”

Laila: “Let’s draw it—like a map of the finger codes.”

Adam: “Great idea! Then we can build the glove to match our drawing.”

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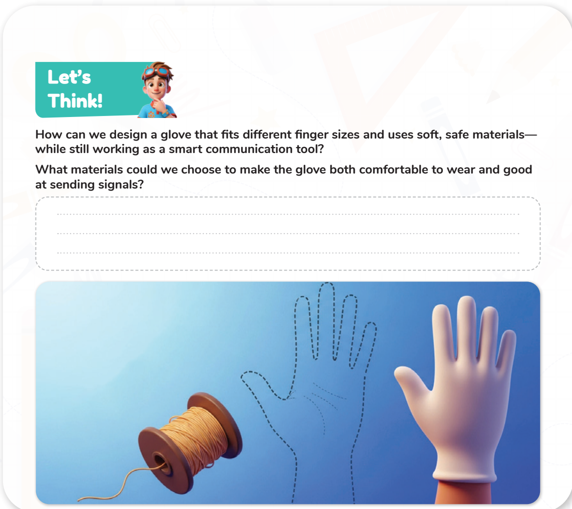

Let’s Think!

How can we design a glove that fits different finger sizes and uses soft, safe materials—while still working as a smart communication tool?

What materials could we choose to make the glove both comfortable to wear and good at sending signals?

Story

Objective:

Students will understand the initial steps in designing a wearable glove circuit, focusing on how copper tape placement and finger taps work together to send signals. This builds foundational knowledge for planning and sketching their own glove designs.

- Gather students and create a curious, collaborative atmosphere.
- Say:
    - o "Today, we join Adam and Laila as they start designing their smart glove. They need to decide where to put the copper tape so that finger taps send the right signals. Let's listen carefully to see how they plan and think like inventors."
- Read the story aloud expressively, emphasizing the discovery and planning process.
- Pause to ask questions such as:
    - o "Why is it important for the tape to touch when two fingers tap?"
    - o "How do you think each finger tap could send a different message?"
    - o "Why might drawing a map of the finger codes help build the glove?"
- Encourage students to imagine themselves designing and drawing their own glove.
- Invite a few students to share predictions or questions before moving on.

Possible Strategies for Implementation:

1. **Engage Through Storytelling**
    - Use an enthusiastic tone to draw students into the design process story.
    - Use props like paper gloves and copper tape for hands-on connection.
2. **Ask Guiding Questions**
    - Pause and pose reflective questions to deepen understanding and curiosity.
3. **Visual Supports**
    - Show diagrams or examples of finger taps creating signals.
    - Encourage students to visualize how different finger pairs can send unique messages.
4. **Relate to Prior Knowledge**
    - Connect to earlier lessons on circuits, sensors, and signal transmission.
5. **Support Diverse Learners**
    - Provide sentence starters and vocabulary support related to circuit design.
    - Use drawings or gestures to aid comprehension.
6. **Encourage Participation**
    - Invite students to share their own ideas about where tape could go and why.

- Facilitate respectful discussion and idea exchange

Let's Think!

Objective:

Students will explore design considerations for wearable technology, focusing on selecting soft, safe, and functional materials to create a comfortable and effective smart glove.

Answers:

- "Soft materials like cotton or knit fabric would make the glove comfortable to wear."
- "Conductive materials like copper tape or conductive thread can carry electrical signals."
- "Bandages or flexible plastic could protect the circuits while still allowing movement."
- "Materials should be safe for skin contact and durable enough for repeated use."
- "The glove needs to fit different finger sizes, so stretchy fabric or adjustable parts could help."

Possible Strategies for Implementation:

1. **Encourage Brainstorming**
   - Prompt students to think about different materials used in gloves and wearable devices.
   - Encourage listing properties such as softness, flexibility, conductivity, and durability.
2. **Facilitate Group Discussion**
   - Organize students into pairs or small groups to share ideas and compare material choices.
   - Encourage respectful listening and building on peers' suggestions.
3. **Use Visual Supports**
   - Show pictures or physical samples of materials like fabric, copper tape, conductive thread, and bandages.
   - Discuss pros and cons of each material for comfort and signal transmission.
4. **Connect to Real-World Examples**
   - Highlight examples of wearable tech materials in smart gloves, fitness trackers, or medical devices.
5. **Support Diverse Learners**
   - Provide sentence starters such as "I think a good material would be... because..."
   - Allow responses through drawing or verbal explanations.

Explorer's Toolkit

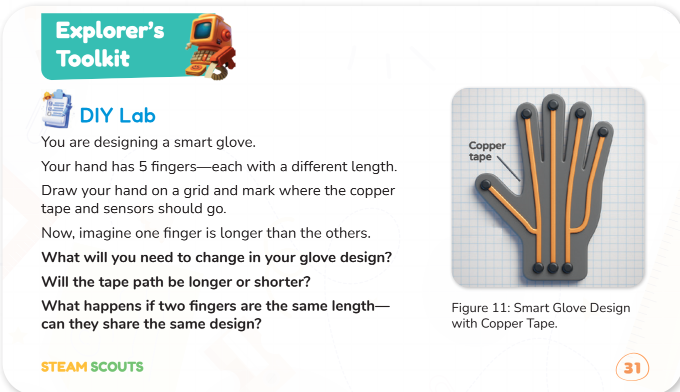
Teacher's Role:

- Introduce the activity by explaining how the coordinate grid helps in planning the glove design.
- Show the example diagram illustrating where copper tape and sensors should be placed on the fingers.
- Guide students in measuring their own finger lengths and labeling them accurately.
- Provide clear instructions for drawing their glove layout on the grid, emphasizing precision.
- Prompt students to think about design changes needed if finger lengths vary or are the same.
- Circulate to support students with measurement, drawing, and spatial challenges.
- Encourage students to explain their design choices and reasoning.

Expected from Students:

- Measure and record the length of each finger accurately in centimeters.
- Draw a clear glove layout on the grid, marking where copper tape and sensors will be placed.
- Consider and describe how finger length differences affect the glove design.
- Participate actively by sharing design ideas and listening to peers.
- Use measurement and spatial skills confidently.

Parts of the book included.



Explorer's Toolkit

DIY Lab

You are designing a smart glove.

Your hand has 5 fingers—each with a different length.

Draw your hand on a grid and mark where the copper tape and sensors should go.

Now, imagine one finger is longer than the others.

What will you need to change in your glove design?

Will the tape path be longer or shorter?

What happens if two fingers are the same length—can they share the same design?

Copper tape

Figure 11: Smart Glove Design with Copper Tape.

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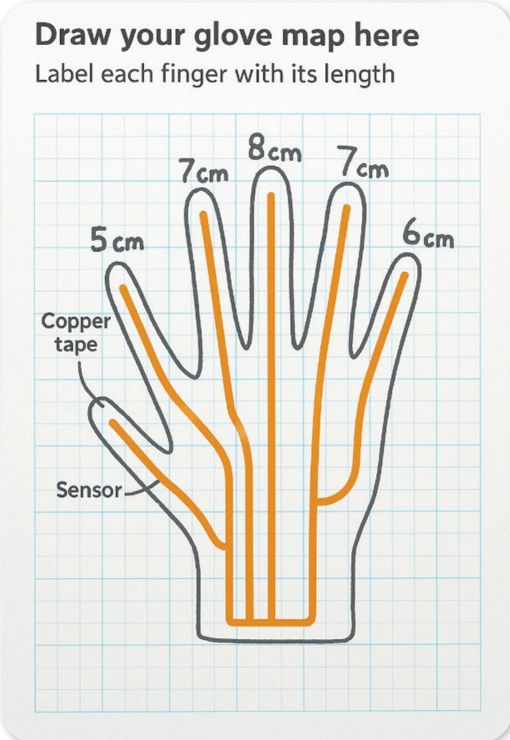
DIY Lab

Objective:

Students will apply coordinate grid concepts to design a smart glove layout, understanding how finger lengths affect sensor and copper tape placement, and practicing spatial reasoning and measurement skills.

Possible Answer:



Possible Strategies for Implementation:

1. **Activate Prior Knowledge**
   - Review basic coordinate grid concepts and measurement units before starting.
   - Use real objects or the students' own hands to demonstrate grid plotting.
2. **Demonstrate the Activity**
   - Model how to measure a finger and mark it on the grid.
   - Show the example diagram and explain the logic behind sensor placement.
3. **Guided Practice**
   - Allow students to measure their fingers with rulers or measuring tapes.
   - Assist with converting measurements to grid units if necessary.
4. **Encourage Critical Thinking**
   - Ask students how they would adjust their design if one finger is longer or if two fingers have the same length.
   - Facilitate group discussions about challenges and solutions.
5. **Support Diverse Learners**
   - Provide printable grid sheets or templates for students who need them.
   - Offer one-on-one help with measuring or drawing as needed.
6. **Visual and Verbal Sharing**
   - Have students present their glove designs to peers, explaining their choices.
   - Use peer feedback to refine ideas.

Explain

Teacher's Role:

- **Provide Clear Instructions and Guidance:** The teacher will explain the concepts of wearable technology, discussing both the technical and creative aspects of designing wearable devices. The teacher will introduce students to the resources they need, including the video link, and guide them through the steps involved in the design and prototype process.
- **Model Thinking:** The teacher will demonstrate how to break down a wearable tech project, including how to identify key components (e.g., sensors, power sources, materials) and their integration. The teacher will model how to approach brainstorming ideas, sketching, and considering the functionality of wearable devices.
- **Facilitate Discussion:** During the "Explain" phase, the teacher will encourage students to share their thoughts and ideas about wearable technology. They will ask guiding questions to help students think critically about their designs and the technology's applications.

Expected from students:

- **Active Engagement:** Students should engage with the video and instructions, taking notes on the wearable design concepts discussed. They are expected to participate in the brainstorming session, sharing their ideas and asking questions when needed.
- **Understanding of Design Principles:** Students should demonstrate an understanding of the basic principles of wearable technology, including the selection of materials, components (e.g., sensors, motors), and the logic behind the design.

Parts of the book included.

Screen Lessons

Click the link to discover more helpful tips and ideas for designing wearable projects.

https://youtu.be/uOepNMAYLyI?si=zRCiSDAPJh8qrygs

Screen Lessons

Objective:

Students will explore wearable technology by designing and building wearable projects that incorporate electronic components like sensors and actuators. They will learn how to integrate technology with clothing or accessories to create functional and innovative designs.

Video Link: https://youtu.be/uOepNMAYLyI?si=zRCiSDAPJh8qrygs

Possible Strategies for Implementation:

1. **Introduction to Wearable Tech:**
   - Begin with a discussion of wearable technology and its various applications in everyday life. Show examples of wearable devices like smartwatches, fitness trackers, and health monitoring systems.
2. **Video Exploration:**
   - Have students watch the linked video to learn about the design principles and tools used for wearable projects. Encourage them to take notes on key ideas they find inspiring or useful for their own projects.
3. **Brainstorm and Design:**
   - After watching the video, students will brainstorm their wearable project ideas. They can start sketching their designs and decide which components they will use.
4. **Prototype Building:**
   - Using Tinkercad or physical materials, students will begin to prototype their wearable devices. They will use sensors, lights, or other interactive components to bring their ideas to life.
5. **Group Discussions and Presentations:**
   - Encourage students to share their designs with the class and explain the functionality of their wearable technology. Provide feedback and suggestions for improvement.

Elaborate

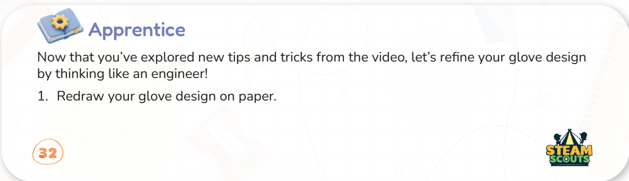
Teacher's Role:

- **Facilitator:** Guide students as they explore wearable technology and its cultural connections. Help them understand how traditional items have influenced modern innovations.
- **Presenter:** Provide examples of wearable technology from different cultures to showcase diverse applications and designs.
- **Questioner:** Ask probing questions to encourage students to think about how wearable technology integrates with cultural traditions, clothing, and lifestyles.
- **Supporter:** Assist students as they explore different wearable technologies and help clarify any confusion about the task. Offer support in research if needed.
- **Collaborator:** Work alongside students as they analyze the role of wearable technology in daily life and help with their design ideas.
- **Assessor:** Evaluate students' understanding by reviewing their reflections on the wearable technology, ensuring they comprehend the impact on daily life.
- **Technology Integrator:** Provide students with the necessary resources (such as digital tools) for research and exploration of wearable technology from different cultures.

Expected from students:

- **Investigative:** Students will research wearable technology from different cultures, paying particular attention to its design, function, and cultural significance.
- **Reflective:** Students will analyze the cultural connections behind wearable technologies and their role in modern life.
- **Collaborative:** Students will discuss their findings in groups and present their ideas to the class.
- **Creative:** Students will be expected to create their own wearable technology concepts, reflecting on how their design can be inspired by a chosen culture's traditions and lifestyle.

Parts of the book included.


Apprentice

Now that you've explored new tips and tricks from the video, let's refine your glove design by thinking like an engineer!

1. Redraw your glove design on paper.

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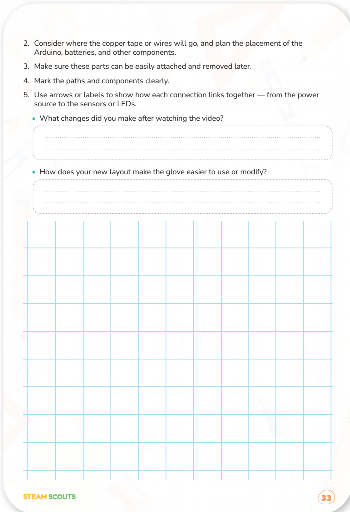
2. Consider where the copper tape or wires will go, and plan the placement of the Arduino, batteries, and other components.
3. Make sure these parts can be easily attached and removed later.
4. Mark the paths and components clearly.
5. Use arrows or labels to show how each connection links together — from the power source to the sensors or LEDs.

- What changes did you make after watching the video?
- How does your new layout make the glove easier to use or modify?

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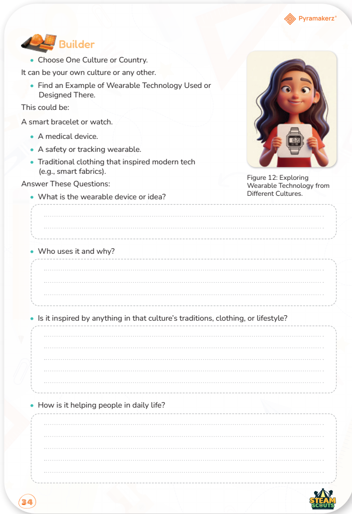
Pyramakerz®

Builder

- Choose One Culture or Country.

It can be your own culture or any other.

- Find an Example of Wearable Technology Used or Designed There.

This could be:

A smart bracelet or watch.

- A medical device.
- A safety or tracking wearable.
- Traditional clothing that inspired modern tech (e.g., smart fabrics).

Answer These Questions:

- What is the wearable device or idea?
- Who uses it and why?
- Is it inspired by anything in that culture's traditions, clothing, or lifestyle?
- How is it helping people in daily life?

Figure 12: Exploring Wearable Technology from Different Cultures.

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Apprentice

Objective:

- Redraw your smart glove design, focusing on the placement of sensors, Arduino, and other components.
- Plan for the wire or copper tape layout, ensuring all components are easily attached and removable.
- Label and mark connections to ensure clarity in how the power and signals are routed to the electric components.
- Refine the layout to make the glove more functional, easier to use, and modifiable.

Answers:

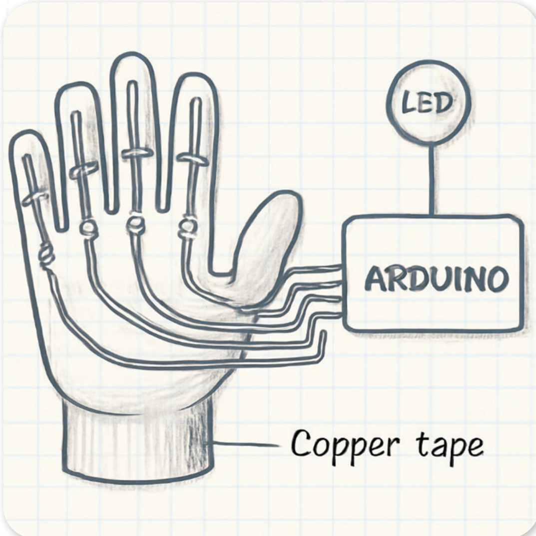
1. What changes did you make after watching the video?

- I modified the design to ensure the placement of components was more accessible. For example, I placed the battery in a more reachable position to make swapping it easier. I also adjusted the position of the sensors for better connectivity and reduced clutter.

2. **How does your new layout make the glove easier to use or modify?**

- The new layout makes it easier to use by simplifying the wiring paths, making them more organized and less likely to tangle. The sensors and Arduino are placed in a way that allows for easier adjustment or replacement without disturbing the other components. The overall design is more modular, so if I need to change a part, I can do so without dismantling the entire glove.

Activity Instructions:



1. Redraw your glove design on paper.
2. Consider the placement of the copper tape, wires, Arduino, batteries, and other components, ensuring they can be easily attached and removed later.
3. Mark the paths and components clearly, using arrows or labels to show how each connection links together (from the power source to the sensors or LEDs).
4. Think about making your design neat and functional, considering how the layout can make the glove easier to use or modify.
5. Reflect on your design by answering the following questions:

- **What changes did you make after watching the video?**
- **How does your new** layout make the glove easier to use or modify?

Possible Strategies for Implementation:

- **Think-Pair-Share:** Allow students to think through their glove layout, discuss it with a partner, and then share their ideas with the class.
- **Guided Reflection:** After students complete their designs, facilitate a short reflection session where they discuss what worked, what didn't, and how they can further refine their design.
- **Peer Review:** Have students exchange their sketches with classmates to get feedback on their designs and make improvements based on peer suggestions.

Builder

Objective:

Students will investigate wearable technology from different cultures or countries, understanding how cultural traditions influence design and how technology meets real-life needs.

Answers:

1. **What is the wearable device or idea?**

- **Example:** A smart bracelet that monitors heart rate and sends alerts.

2. **Who uses it and why?**
   - **Example:** Used by elderly people or athletes to track health and prevent emergencies.
3. **Is it inspired by anything in that culture's traditions, clothing, or lifestyle?**
   - **Example:** Inspired by traditional jewelry designs unique to the culture, blending fashion with technology.
4. **How is it helping people in daily life?**
   - **Example:** Helps users monitor health easily, stay safe, and remain connected to caregivers

Possible Strategies for Implementation:

1. **Clearly Explain the Task and Objectives.**
   - Begin by discussing the importance of wearable technology and how it varies worldwide.
   - Explain that students will explore a culture or country's wearable tech, focusing on its purpose, cultural inspiration, and impact.
2. **Provide Research Guidance and Resources.**
   - Teach students how to identify trustworthy sources online, in books, or through videos.
   - Share curated lists of websites, articles, or videos related to wearable technology and cultural innovation.
   - Encourage note-taking strategies that focus on answering the activity's key questions.
3. **Model the Research Process**
   - Demonstrate how to search for information using specific keywords.
   - Show how to extract relevant facts related to the wearable device, user, cultural influence, and daily life benefits.
4. **Encourage Critical and Cultural Thinking.**
   - Prompt students to consider how culture shapes design choices, materials, and technology use.
   - Discuss examples of traditional clothing or customs influencing modern wearable tech.
5. **Facilitate Collaborative Learning and Sharing.**
   - Allow students to work individually or in pairs to support different learning preferences.
   - Organize presentations or small group discussions for students to share findings and learn from peers.
6. **Use Scaffolds and Differentiated Supports.**
   - Provide graphic organizers or guided question sheets to help structure research and responses.

- Offer sentence starters for students who need help articulating their thoughts in writing.
- Encourage alternative formats for presenting information, such as posters, slideshows, or videos.

7. **Connect to Real-World Contexts.**
   - Relate student research to local or familiar cultural examples when possible.
   - Highlight how wearable technology can solve everyday problems and enhance lifestyles globally.

Evaluate

Teacher's Role:

- **Facilitator of Hands-On Learning**
  - Support students during the cutting and construction process. Encourage precision and care in matching finger lengths to their grid sketches.
  - Offer help if students struggle with fabric handling or glove alignment.
- **Encourager of Iteration**
  - Remind students that mistakes are part of the design process. If the glove doesn't fit perfectly, it's an opportunity to reflect and improve.
- **Ask prompting questions:**
  - "How can you adjust the next version?"
  - "What would you change in your pattern?"
- **Leader of Constructive Feedback**
  - Guide students during the peer review step. Help them give kind, specific, and helpful feedback.
  - Reinforce respectful communication when trying on each other's designs and evaluating comfort and usability.

Expected from Students:

- **Test and Reflect on Fit**
  - Try on their gloves and evaluate if the fit is accurate and comfortable.
  - Consider questions like "Does each finger fit?" and "Can I move my hand easily?"
- **Revise if Needed**
  - Make adjustments if the glove is too tight, loose, or misaligned.
  - Embrace the concept of "design–test–improve" as part of the engineering mindset.
- **Conduct Peer Review**
  - Swap gloves with a partner and test each other's glove designs.

   - Use the guided questions to give thoughtful feedback on comfort, flexibility, and areas for improvement.
- **Practice Design Communication**
   - Explain clearly what worked and what didn't.
   - Respectfully receive feedback and consider how to implement it.

Showcase

Objective:

Students will apply their design skills by creating a physical prototype of their smart glove, testing its fit and functionality, and engaging in peer feedback to refine their designs through iterative improvement.

Answer:

1. **How does it feel?**
   - Students might describe sensations such as "comfortable," "a bit tight around the knuckles," or "fingers move freely."
2. **Is it easy to move your fingers?**
   - Expected answers include "Yes, the design allows smooth movement," or "No, it feels stiff in some areas."
3. **What worked well in their design?**
   - **Examples:** "Good finger length measurement," "Flexible fabric choice," "Accurate sensor placement."
4. **What could be better?**
   - **Examples:** "Make the wrist opening wider," "Add more fabric between fingers," "Adjust the length of the index finger."

Step-by-Step Instructions:

1. **Prepare Materials:**
   - Gather fabric, scissors, rulers, and markers for cutting and marking the glove.
   - Use the student's glove sketch from the grid as a template for cutting.
2. **Cut the Glove Shape:**
   - Carefully cut the fabric following the glove sketch outline.
   - Ensure each finger length on the glove matches the student's actual finger length for a proper fit.
3. **Assemble and Wear:**
   - Put the glove on and check for comfort and fit on each finger.
   - Note any areas that feel tight, loose, or restrictive.

4. **Adjust and Improve:**
    - If the glove is too tight or loose, make adjustments by trimming or reshaping the fabric.
    - Re-test the fit after each modification.
5. **Peer Swap and Feedback:**
    - Exchange gloves with a classmate.
    - Try on each other's gloves and answer the guided questions about comfort, finger movement, and design effectiveness.
6. **Reflect on Design:**
    - Discuss what worked well in the design and what could be improved.
    - Consider peer feedback for further refinements.

Possible Strategies for Implementation:

1. **Demonstrate the Cutting Process:**
    - Show students how to transfer their sketch to fabric and cut carefully.
    - Emphasize safety with scissors and precision in cutting.
2. **Model the Fitting and Adjustment Cycle:**
    - Demonstrate trying on the glove and identifying fit issues.
    - Show how to make simple adjustments to improve fit.
3. **Encourage a Growth Mindset:**
    - Reinforce that design is iterative; initial attempts may not be perfect, and improvement is part of the process.
    - Celebrate persistence and problem-solving.
4. **Facilitate Effective Peer Feedback:**
    - Teach students how to give constructive, respectful feedback focusing on specific aspects like comfort and movement.
    - Provide sentence starters or a feedback checklist to guide comments.
5. **Support Diverse Learners:**
    - Provide templates or pre-cut fabric pieces if needed.
    - Allow verbal or visual feedback options for students who find writing challenging.

Rubric: Smart Glove Prototype.

Criteria	Excellent (4 Point)	Good (3 Points)	Fair (2 Points)	Needs Work (1 Point)
Accuracy of Glove Design	Glove matches the grid design precisely with correct finger lengths and sensor placements.	Glove mostly matches design with minor inaccuracies in length or placement.	Glove shows some mismatch with design; finger lengths or placements are inconsistent.	Glove design does not reflect the grid plan; many inaccuracies.
Fit and Comfort	Glove fits comfortably; allows full finger movement without restriction.	Glove fits fairly well; minor discomfort or slight movement restriction.	Glove fits loosely or tightly; noticeable discomfort or limited movement.	Glove does not fit properly; uncomfortable and restricts movement significantly.
Adjustment and Iteration	Student made thoughtful adjustments based on testing and peer feedback.	Student made some adjustments to improve fit/design.	Student made minimal or ineffective adjustments.	No adjustments made despite issues.
Peer Feedback and Reflection	Provides detailed, constructive feedback and thoughtfully reflects on own and peer designs.	Provides helpful feedback and reflects adequately on designs.	Feedback or reflection is vague or incomplete.	Feedback and reflection are missing or superficial.
Creativity and Effort	Demonstrates creativity in design and shows high effort and engagement.	Shows some creativity and good effort.	Limited creativity or inconsistent effort shown.	Little creativity or effort demonstrated.

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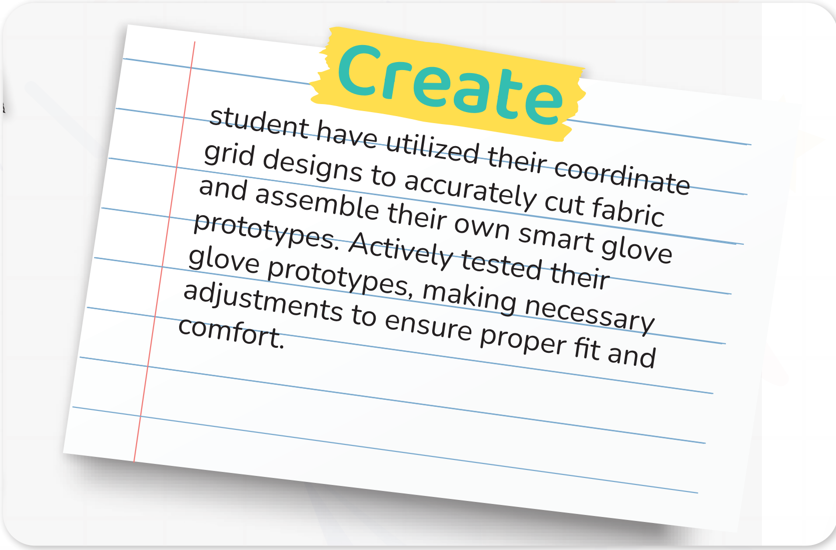

Now I Can

Objective:

Students will reflect on their learning and identify how they've grown in understanding communication, empathy, and the role of assistive tools in helping others share ideas.

Activity Instructions:

1. **Introduce the Reflection**
   - Say: "Let's look back on everything we've explored this lesson. You learned how it feels to face communication challenges, and how tools can help people share their thoughts in special ways."
2. **Guide Students Through the Checklist**
   - Read each statement aloud one at a time.
   - After each one, ask students to give a thumbs up, sideways, or down to show how confident they feel.
   - Example prompts:
     - "Did you learn how people share ideas without talking?"
     - "Do you know how tools can help people talk?"
3. **Student Completion**
   - Let students check off or color the "Now I Can" boxes they feel good about.
   - Remind them: "It's okay if you didn't check all of them. We're all still learning."
4. **Optional Extension**
   - Ask students to circle the one they're most proud of.
   - Invite them to write a sentence or draw a picture to show their favorite learning moment.

Relation to Standards

1. NGSS (Next Generation Science Standards).

- MS-ETS1-1: Define problems that can be solved through engineering design.
- MS-ETS1-2: Evaluate competing design solutions based on criteria and constraints.
- MS-ETS1-3: Analyze data from tests to determine design improvements.

2. CCSS (Common Core State Standards).

- CCSS.MATH.CONTENT.6.NS.C.6: Understand and use positive and negative numbers on a coordinate plane.
- CCSS.ELA-LITERACY.SL.6.1: Engage effectively in collaborative discussions with diverse partners.

3. ISTE (International Society for Technology in Education).

- Empowered Learner: Students use technology and design thinking to solve problems.
- Knowledge Constructor: Students critically curate resources and use digital tools for learning.